

Self-organisation of functional cortical maps without macroscopic spatial patterning

Short title: Self-organised maps without macroscopic spatial patterning

Nicola Mendini^{1,*}, Stuart P. Wilson¹

¹ School of Computer Science, The University of Sheffield, Sheffield, United Kingdom

* nmendini1@sheffield.ac.uk

Abstract

Self-organisation, driven by correlational structure in early sensory experience, is a compelling theory of cortical map development. Topological maps for visual orientation preference, for example, emerge in simulations where cortical units interact laterally via short-range excitation and long-range inhibition, and connection weights adapt through Hebbian learning. These maps closely resemble those measured in the primary visual cortex (V1) of many species, with similar orientation preferences encircling ‘pinwheel’ discontinuities, and orientation-selective connections over large distances. An important challenge to this theory is that the cortices of some species have similar functional properties, but display no spatial patterning, such as the ‘salt-and-pepper’ organisation of visual orientation preferences in rodent V1. Here we ask whether maps with matching functional properties can self-organise in a model of visual cortex development, in a parameter regime that does not give rise to detectable macroscopic spatial patterning. We show that when lateral interactions are very short range, populations of neuronal receptive fields can emerge that can encode visual stimuli with comparable fidelity and complexity, while producing patterns of orientation preference that appear random under sensible assumptions of cellular displacement in the cortical tissue. However, an additional longer-ranging excitatory interaction between cortical units was required to also ensure higher levels of noise robustness in cortical dynamics. The resulting model demonstrates how salt-and-pepper maps can self-organise by the

same basic principles as topological maps, and encode visual stimuli with comparable efficiency. As such, specific spatial patterns in the arrangement of receptive fields on the tissue may be thought of as signatures of the different selection pressures that have been exerted on a common underlying developmental process.

Author summary

In the visual cortex of many mammals, neurons preferring similar visual features form smooth spatial maps. In others, notably rodents, equally selective neurons appear intermingled in a seemingly random “salt-and-pepper” pattern. Does this difference require distinct developmental mechanisms, or can both forms of organisation emerge from related principles?

Using a computational model of visual-cortex development, we show that competitive interactions and experience-dependent learning can produce functional organisation at very different spatial scales. Large interaction neighbourhoods generate the familiar extended domains of cortical maps. When interactions operate at finer scales, the model instead forms many small, orderly domains whose spatial pattern can be obscured by modest positional disorder. These fine-scale networks preserve stimulus information and representational structure, while an additional selectively learned excitatory interaction restores robust cortical dynamics.

Our results show that the absence of a visible macroscopic map does not necessarily imply the absence of developmental self-organisation. They raise the possibility that salt-and-pepper cortex contains fine-scale structure that is difficult to detect from neuronal positions alone but remains evident in connectivity and population activity. More broadly, cortical map geometry may reflect different spatial and wiring regimes of a common developmental process.

Introduction

Stimulus-driven self-organisation is a successful theory of the development of neocortical maps, as represented by computational models that reproduce realistic neuronal receptive fields and spatial patterns of receptive fields across simulated cortical areas.

Accordingly, correlated sensory inputs, Hebbian plasticity, and competitive recurrent interactions between cortical neurons turn emergent neuronal responses to input stimuli into stable activity patterns that persist across developmental time [1–4]. Competition is typically implemented by a Mexican-hat pattern of lateral interactions—short-range excitation and longer-range inhibition—which over time consolidates Hebbian reinforcement of correlated activity into macroscopic spatial patterning across the cortical sheet [4, 5] (see Fig 1).

Fig 1. Classical self-organisation in GCAL. (a) LGN input drives a recurrent V1 sheet whose lateral interactions have a Mexican-hat form: short-range excitation (SRE) and longer-range inhibition (LRI). During settling, these interactions shape stimulus-evoked activity; over development, Hebbian plasticity consolidates the correlations into a continuous orientation map with extended iso-orientation domains and pinwheel centres. The characteristic domain spacing is set by the spatial scale of the lateral interactions. (b) The two main geometric variables are the total number of cortical units, N_{ctx} , and the effective number within the short-range excitatory neighbourhood, N_{lat} .

In species that display clear topological maps, the spatial patterning is accompanied by structured intrinsic connectivity. Long-range horizontal projections form patches across the cortical surface, preferentially linking neurons with similar response properties, often with a spacing comparable to that between functional domains [6–14]. Self-organisation therefore provides a way to understand both the spatial patterns of receptive field organisation across a cortical area, and the motifs of recurrent wiring that form between like-tuned neurons, as manifestations of a common underlying developmental process [15, 16].

The emergence of orientation preference maps in primate primary visual cortex (V1) is a well-studied example. Adjacent neurons tend to respond selectively to similar orientations of visual edge, regions of the tissue in which neurons have similar orientation preference form extended iso-orientation domains, and these domains are arranged continuously around point discontinuities referred to as pinwheels [17–22]. Self-organising models also readily account for several other features of neocortical organisation, including the observation that the density of pinwheel centres, in units of the typical spacing between iso-orientation domains, converges towards π across many species [23–26].

The challenge for self-organisation as a general theory of cortical development is that

not all functional organisation in cortex presents with such macroscopic spatial patterning. In V1 of rodents and lagomorphs, orientation preferences appear randomly arranged across the tissue, with nearby neurons corresponding to unrelated orientations and an absence of extended iso-orientation domains or pinwheels [27].

Importantly, the apparent lack of spatial patterning in the distribution of receptive fields in rodent V1, often referred to as *salt-and-pepper*, does not correspond to an absence of functional organisation. Salt-and-pepper cortex still presents sharply tuned neurons and reliable neuronal responses, as well as some measurable local structure, such as micro-clustering and tuning correlations over very short distances [28–32]. Similarly, although rodents often lack the patchy long-range horizontal connectivity associated with topological maps, local connectivity between cortical neurons may not be random, instead exhibiting elongation along the retinotopic axis that corresponds with their preferred orientations [33, 34].

According to computational models of self-organisation, the competitive interactions between neurons that determine the distribution of receptive fields in a cortical area are also responsible for the emergence of a spatial pattern. Hence, functional organisation and spatial patterning are inextricably linked in topological maps. But if some cortices exhibit functional organisation without displaying clear spatial patterns, it becomes important to ask whether spatial patterns are indeed an essential consequence of those competitive interactions, or only one possible regime of self-organisation.

In other words, can competitive Hebbian learning generate useful functional organisation without also producing the kind of macroscopic map pattern by which self-organisation is usually recognised?

If so, rodent neocortical development may not reflect a failure of the self-organising process per se, e.g., due to differences in afferent sampling, retino-cortical geometry, or additional sources of randomness in the afferent projection [35–43], but reflect instead the outcome of a qualitatively different form of developmental self-organisation.

Here we investigate whether a different regime of functional map self-organisation may exist for a different spatial scale of competitive lateral interactions, without necessarily yielding a clear macroscopic spatial signature. First, we use a canonical self-organising model of macro-scale orientation map development (GCAL; [4]) to determine whether reducing the range of lateral interactions is sufficient to preserve

functional map properties without producing spatial patterns. We show that as the scale of competitive lateral interactions is reduced, robustness of neuronal dynamics to noise diminishes, but that noise robustness can be recovered at small spatial scales by the introduction of an additional long-range stabilising interaction. In this new regime of self-organisation, we find that spatial patterning may appear random, suggesting that rodent neocortical maps reflect a qualitatively different solution to the development of cortical function.

Results

Evaluating cortical maps

The crucial factor that determines the length scale of topological maps produced by self-organising models of cortical development is the range of lateral interactions between cortical neurons, under the constraint that those interactions are net excitatory over short distances and net inhibitory over longer distances. It is plausible that salt-and-pepper maps may be inherently similar in structure to topological maps, but self-organise at a length scale that is so small that patterning becomes undetectable once cells have finished growing into the tissue.

To explore this possibility, we allowed primary visual cortex maps to self-organise in simulation using the GCAL computational model of primary visual cortex (V1) development [4], varying i) the size of V1 and ii) the size of a neuron's lateral interaction neighbourhood between simulations.

In GCAL, map development is modelled by projecting a stimulus (e.g., an image) via fixed connection weights to a 2D neural sheet representing the thalamus (e.g., the LGN), then projecting thalamic activity via plastic connection weights to a second sheet representing a cortical field (e.g., V1), and allowing cortical activity to settle through fixed short-range excitatory and plastic long-range inhibitory recurrent connections, before adjusting all plastic weights via Hebbian learning, and repeating many times with randomly drawn stimuli.

The first independent variable, the size of V1, was thus parametrised by N_{ctx} , the total number of cortical units in the simulated cortical sheet, noting that the afferent

input sheet was fixed at size $N_{\text{Ign}} = 24^2$, so that $N_{\text{ctx}} = \Delta N_{\text{Ign}}$ defines Δ as the cortical expansion factor. The second independent variable, the size of a neuron’s lateral interaction neighbourhood, was parametrised by N_{lat} , the number of other cortical units that each interacts with laterally via short-range excitatory connections, while longer-range lateral inhibitory interactions were always restricted to a distance three times that of excitation.

After each cortical map had self-organised, we measured three functional properties of the visual stimulus encoding—the *fidelity*, *complexity*, and *robustness* to noise—following efficient-coding and information-bottleneck principles [44–46]. If spatial patterns and functional mappings can be independent emergent properties of cortical development, we might expect these measures to be largely unaffected by the length scale at which a mapping self-organises, even at shorter length scales where spatial patterns are expected to increasingly resemble rodent salt-and-pepper organisation.

The *fidelity* of a cortical mapping represents the amount of information about an afferent input pattern that persists after the recurrent cortical interactions that are elicited by that stimulus have settled. A practical way to capture this is to measure the accuracy with which afferent (LGN) stimuli can be decoded from the corresponding activity patterns in V1 over a dataset of natural image stimuli. To this end, we can use a simple nonlinear decoder and measure the average cosine similarity A between the afferent patterns reconstructed from V1 using that decoder and the actual afferent stimulus patterns (Fig 2a). To obtain a normalised and unitless measure between 0 and 1, we compare this average reconstruction accuracy with a baseline A_{null} obtained when reconstructing the input patterns from non-informative, constant V1 activity, resulting in a measure of fidelity $(A - A_{\text{null}})/(1 - A_{\text{null}})$.

One simple way to estimate the *complexity* of a cortical mapping is to generate a dataset of stimulus-evoked activity patterns and determine for that dataset the number of principal components that are required to explain a given threshold level of the variance amongst them; a mapping for which fewer principal components are required to explain 95% of the variance, for example, should be considered less complex. To obtain a normalised and unitless measure of complexity, we perform this analysis on the stimulus-evoked activity patterns at the level of the thalamus to obtain D_{Ign} , and again at the level of the cortex, after recurrent settling, to obtain D_{ctx} , and express complexity

Fig 2. A fidelity–complexity trade-off and a shared functional reference. (a) Fidelity is measured by decoding LGN activity from settled V1 activity, calculating the mean cosine similarity between true and reconstructed test presentations, and adjusting this value relative to the null-decoder baseline. (b) Representational complexity is the number of PCA components required to explain 95% of response variance, reported as $D_{\text{ctx}}/D_{\text{lgn}}$, with $D_{\text{lgn}} = 87$ under valid random cropping. (c–d) Increasing lateral-neighbourhood size decreases fidelity and complexity at expansion factors $\Delta = N_{\text{ctx}}/N_{\text{lgn}} \in \{1, 4, 9\}$. Points and error bars show the mean and ± 2 standard deviations across three independent runs. (e) For each sheet size, neighbourhood size was eliminated analytically between the fitted fidelity and complexity functions; the three derived curves were averaged pointwise. The dashed reference is the saturation-side maximum-curvature point C_{curv} , which occurred near $D_{\text{ctx}}/D_{\text{lgn}} = 0.5$ and corresponded to $R_{\text{macro}}^* \approx 72$.

through the relative dimensionality $D_{\text{ctx}}/D_{\text{lgn}}$ (see Fig 2b). Where $D_{\text{ctx}}/D_{\text{lgn}} < 1$, the V1 settling process has reduced the complexity of the stimulus representation.

To determine the *robustness* of each cortical mapping, we add temporally correlated but spatially independent noise to the drive of each cortical unit at every step of recurrent settling, and compare the resulting settled cortical activity pattern with that obtained for the same stimulus in the absence of noise. Robustness is measured as the noise intensity at which the similarity between these activity patterns, averaged across stimuli, falls to 0.95.

Against these criteria we can now define a functional cortical mapping to be one of high fidelity, appropriately reduced complexity, and high robustness to noise, and explore how these factors vary with the parameters that determine the length scale of cortical self-organisation.

Scaling down classical maps preserves coding but weakens robustness

The GCAL model was simulated across a wide range of lateral interaction neighbourhood sizes, from extremely local interactions involving only nearby units towards $N_{\text{lat}} \approx 80$, using small, medium, and large V1 sheets, defined by V1/LGN side-length ratios of 1, 2, and 3, respectively, corresponding to $\Delta = 1, 4$, and 9. At all three V1 sizes, we found that self-organisation via larger interaction neighbourhoods reduces the complexity, but also the fidelity, of cortical mappings (Fig 2c–d).

Interestingly, map fidelity and complexity did not vary independently with V1 size

and lateral neighbourhood size. Instead, fidelity increased monotonically with
complexity, and the results from all three sheet sizes fell naturally along the same
fidelity–complexity trade-off curve (Fig 2e). The position of a cortical map on this
trade-off was determined by the size of the recurrent neighbourhood relative to the size
of the cortex, expressed as the ratio $R = N_{\text{ctx}}/N_{\text{lat}}$.

To identify a common reference point on this trade-off, we fitted fidelity and
complexity against lateral-neighbourhood size for each cortical sheet size. Each pair of
fits described how fidelity varied with complexity at that sheet size. Because the three
relationships closely overlapped, we averaged them to obtain a shared curve.

We used the point where the shared curve bends most strongly towards its plateau
as a consistent reference for comparing maps. This point, denoted C_{curv} , occurred at a
complexity of approximately 0.5.

Mapping the knee of the shared curve back through the sheet-size-specific fits
yielded a common size ratio $R^* = N_{\text{ctx}}/N_{\text{lat}} \approx 72$. Therefore, setting the size of the
cortex N_{ctx} to approximately R^*N_{lat} allowed maps of different spatial scales to be
compared at the same reference point on the fidelity–complexity relationship.

To explore further, we ran a new set of simulations (Fig 3), varying N_{lat} and setting
 N_{ctx} so as to preserve the ratio $N_{\text{ctx}}/N_{\text{lat}} = R^*$ found at the shared reference point. We
verified that complexity and fidelity levels remained comparable. However, we found
that noise robustness increased monotonically with the size of the lateral neighbourhood
(Fig 3d). Large lateral neighbourhoods produced cortical maps that had topological
patterns and were highly robust to noise, whereas small lateral neighbourhoods
produced maps with apparently random patterns that were instead highly sensitive to
noise.

Fig 3. Robustness increases as matched classical maps are scaled up. (a)
Example of recurrent settling for a fixed stimulus (left). Adding temporally correlated
noise to the pre-nonlinearity V1 drive during settling perturbs the final state (right).
We quantify stability as the cosine similarity between the settled state without noise
and the settled state with noise for the same input stimulus, parametrised by the
additive noise intensity σ_{noise} , and robustness as the value of noise intensity that makes
stability drop to 0.95. (b) Example single-neuron traces with and without noise
illustrate the divergence in settled activity. (c) Stability decreases with noise intensity
 σ_{noise} , and the rate of degradation depends on scale when models are matched near
 $R^* = N_{\text{ctx}}/N_{\text{lat}} \approx 72$. Larger sheets and lateral neighbourhoods are more robust when
the two variables are increased together at fixed R^* . (d) Robustness increases
monotonically with N_{lat} over the tested range.

Selective excitation restores robustness at fine scale

Our results with GCAL as a model of cortical development suggest that as long as cortical size and neighbourhood size are appropriately balanced, cortical maps with widely varying spatial patterns can self-organise to represent stimuli with comparable fidelity and complexity, but only topological maps can also be robust. We therefore lack a clear picture of why nature may have selected, or at least tolerated, non-topological maps in some species. Might an extension of the GCAL model to specifically increase noise robustness also preserve high fidelity and low complexity in the non-topological patterning regime?

To explore this possibility we continued the investigation using an extension of the GCAL model to include an additional longer-range lateral excitatory interaction between cortical units that might help stabilise the recurrent dynamics and ultimately improve noise robustness (Fig 4). We will hereafter refer to the original GCAL model as macro-GCAL and the extended model as micro-GCAL for clarity. As in macro-GCAL, the short-range excitatory neighbourhood defined the lateral interaction scale.

Fig 4. Micro-GCAL, an interaction motif for fine-scale cortical organisation. Starting from the same developmental ingredients as GCAL, the full motif retains short-range excitation (SRE) and longer-range inhibition (LRI), and adds cross-domain excitation (CDE). CDE is a selectively learned excitatory interaction with the widest spatial reach ($r_{\text{sre}} < r_{\text{lri}} < r_{\text{cde}}$). Together with the regime-specific radii and interaction balance, the motif redistributes coordination from a small number of large domains to many smaller coupled domains.

In micro-GCAL, we fixed the spatial ranges of longer-range inhibition and cross-domain excitation as multiples of the short-range excitatory neighbourhood, and varied the size of this neighbourhood to match the fidelity and complexity achieved by macro-GCAL at its shared reference point. We denote the corresponding reference ratios by R_{macro}^* and R_{micro}^* , which were approximately 72 and 523, respectively. Holding $R = N_{\text{ctx}}/N_{\text{lat}}$ near R_{micro}^* while increasing N_{lat} again produced a monotonic increase in noise robustness, reaching values comparable with macro-GCAL (Fig 5d). However, even at the largest values of N_{lat} , where the cortical maps were most robust, micro-GCAL produced much finer spatial patterns rather than macro-domain maps.

The emergent organisation of the cortical maps produced by the micro-GCAL model reveals a population of similarly well-tuned receptive fields, which cover the orientation

Fig 5. Micro-GCAL matches macro-GCAL fidelity, representational complexity, and settling robustness through CDE. (a) Example PCA spectrum (first 15 components) for micro-GCAL, used to compute D_{ctx} . (b) Micro-GCAL was matched to the fidelity and complexity achieved by macro-GCAL at its shared reference point, $D_{\text{ctx}}/D_{\text{ign}} \approx 0.5$. The two outliers reflect the smallest possible solutions in the tested range, where convergence is not as stable. (c) Contribution of CDE to noise stability under matched fine-scale geometry and gain-calibration targets. In the CDE-absent condition, recurrent excitation was confined to SRE; in full micro-GCAL, excitation was redistributed between SRE and CDE. Stability increased with the addition of CDE at the comparison shown. (d) Micro-GCAL robustness shows a scale dependence on N_{lat} similar to macro-GCAL over the range in which its domain spacing is resolved on the lattice. The reference curve from macro-GCAL is rescaled by the factor $R_{\text{macro}}^*/R_{\text{micro}}^* = 72/523 \approx 0.138$ to allow comparison with micro-GCAL. Smaller neighbourhoods are omitted when the matched micro-domain period would fall below lattice resolution.

domain uniformly and produce a globally smooth retinotopy, with highly structured
 learnt lateral interaction weights (see Fig 6). The difference between these maps and
 those produced by macro-GCAL is only the scale of the spatial pattern, with the
 organisation now distributed across a large number of very small domains that are
 locally coupled, rather than a small number of larger domains. Therefore, micro-GCAL
 does not simply produce a noisy or degraded version of a classical map. On the
 idealised lattice of a simulated cortical sheet, the organisation is equally clear and
 orderly, but it is expressed at a finer spatial scale.

Fig 6. Example micro-domain map produced by micro-GCAL. (a) Map outputs. (i) A developed map with many small, locally interconnected domains, with an example spacing of $\Lambda \approx 4$ model units. (ii) Feedforward receptive-field (RF) weights for one sampled neuron and nearby units. (iii) A retinotopic “fishnet” remains globally smooth with local, domain-scale distortions. (iv) The orientation-preference histogram is approximately flat. (b) Lateral structure and responses. (i) Lateral weight profile from the sampled neuron to all other neurons shows domain-scale structure and elongation aligned to the neuron’s preferred orientation (dashed line). The circle indicates the effective span of cross-domain excitation used during self-organisation. (ii) Examples of settled activity patterns evoked by naturalistic inputs after development. (iii) Mean Fourier amplitude spectrum across many responses, showing the annular band used, together with orientation coverage, to select the reported radius ratios. Together, these receptive-field, retinotopic, coverage, connectivity, and response signatures show the usual convergence of a well-organised self-organising map while its characteristic domain area remains over an order of magnitude smaller than the matched macro-GCAL value.

Small-scale functional maps may appear salt-and-pepper

The ideal lattice makes even very fine organisation easy to see, whereas real neurons are irregularly positioned and their relative locations can change during development. Detectability of any underlying pattern therefore depends on positional disorder relative to the spacing between domains. Tree shrew and gray squirrel provide a useful scale-matched comparison because their V1 areas and visual acuities are broadly comparable, while tree shrew V1 contains clear macro-domain maps and gray squirrel V1 has a salt-and-pepper organisation [11, 42, 47].

We represented a $1\text{ mm} \times 1\text{ mm}$ cortical patch with a 45×45 crop containing 2025 units, close to estimates of approximately 2000 neurons per mm^2 in non-primate cortical surfaces [42]. Macro-GCAL was calibrated so that its domain spacing matched the tree-shrew value of approximately $600\text{ }\mu\text{m}$ [24, 42], corresponding to 27 mean neuronal spacings in the model. We then used the same sheet size and functional reference point to generate the corresponding micro-GCAL map.

After maps had self-organised, we increased local positional displacement until the macro-GCAL map was visually compatible with the experimental tree-shrew map. The selected displacement moved units by an average of about two lattice positions, approximately $50\text{ }\mu\text{m}$ under this calibration. Applying the same displacement distribution to micro-GCAL had a much stronger effect: the macro-domain map remained recognisable, whereas the periodic structure of the micro-domain map disappeared (Fig 7a). The same level of displacement resulted in the exchange of neuronal tuning properties across a much larger fraction of the distance between micro-domains, obscuring the spatial pattern in the micro-GCAL model only.

To determine what quantitative traces of organisation survived, we first examined tuning-based autocorrelation and orientation-difference measures. Once the displaced micro-domain map had lost its visible periodicity, these measures retained structure only over short distances (Fig 7c), consistent with the weak local clustering reported in mouse V1 [30]. We next retained lateral Hebbian weights above a minimum threshold and sampled up to a fourth of the full connection-field positions to approximate sparse anatomical sampling, and applied the analysis used by Van Hooser et al. [33] to search for periodicity (patchiness) in the lateral connectivity profiles. The method recovered

Fig 7. Post-learning positional permutation can mask micro-domain periodicity and patchy connectivity signatures. (a) Illustrative scale-matched comparison. (i) A tree-shrew-scale macro-domain simulation and (ii) a micro-domain simulation with the same sheet size at the matched functional reference point. After applying the same displacement distribution (realised mean ~ 2 lattice spacings, or $d \approx 50 \mu\text{m}$ under the illustrative lattice-to-tissue conversion), macro-domain structure remains visible (iii), whereas micro-domain periodicity becomes difficult to detect (iv). Panels (v–vi) show representative tree-shrew and squirrel maps for qualitative context [11, 47]. (b) Connectivity-based detectability. In an ideal lattice without displacement, micro-domain organisation yields elongated, like-to-like connectivity fields (i) that are detectable using the analysis of Van Hooser et al. [33] (ii), but at high spatial frequency (approximately 6 cycles/mm, compared with approximately 1.7 cycles/mm for tree shrew). After displacement, patchiness becomes very difficult to detect (iii–iv). Experimental reference panels from Van Hooser et al. [33] are shown for comparison (v–vi). (c) Periodicity-based detectability. Radially averaged axial autocorrelation and orientation-preference difference suggest that periodicity could be detectable only when displacement is small relative to micro-domain spacing.

high-frequency periodicity before displacement, but no clear periodic peak remained afterwards (Fig 7b). The remaining fields still preferentially linked similarly tuned units and were elongated along their preferred orientation, yet the sparse connectivity sample no longer revealed any overall spatial pattern. Modest positional displacement therefore transformed an orderly, selective, and robust micro-domain map into one with the visual and quantitative signatures of salt-and-pepper cortex.

Spatially masked micro-domain maps preserve stimulus-related population structure

The displacement experiment showed how spatial patterning could be absent from the cortical surface while the neuronal response properties acquired during development remain. We therefore asked whether micro-GCAL preserved stimulus-related population organisation comparable to macro-GCAL despite their different spatial layouts. Static sinusoidal gratings were used as visual stimuli to evoke population responses in V1, each response was considered a point in a high-dimensional space (one dimension per neuron), and Uniform Manifold Approximation and Projection (UMAP) was used to visualise the topological structure in those stimuli that may have been preserved in those responses.

Full-phase sinusoidal gratings offered a particularly informative stimulus family because their angle-phase relationships have a known geometry. If α denotes wave-vector angle and ϕ spatial phase, reversing the wave vector also reverses phase, so

$(\alpha + \pi, \phi)$ and $(\alpha, -\phi)$ describe the same image. This joins opposite sides of the angle-phase space with a twist and gives the family of distinct gratings a Klein-bottle identification [48]. We projected the grating images and the corresponding settled responses from macro-GCAL and micro-GCAL separately with UMAP.

The stimulus embedding formed a smooth folded surface with an orderly progression of grating angle (Fig 8a). Despite the very different appearance of their respective maps, the macro-GCAL and micro-GCAL response embeddings closely reproduced this broad organisation, including the same smooth angle progression across similarly folded surfaces (Fig 8b–c). Their response geometry was therefore compatible with the Klein-bottle structure of the stimulus family and remained consistent across the two spatial regimes.

Fig 8. Population-response embeddings preserve grating-angle organisation across spatial regimes. (a) Example gratings and a UMAP projection of the stimulus set, coloured by grating angle. (b–c) Settled macro-GCAL and micro-GCAL responses to the same gratings, with separately fitted UMAP projections. Both form smooth folded surfaces with an orderly angle progression despite their different cortical layouts. (d) A selected high-population-activity subset of mouse V1 responses to oriented stimuli from Stringer et al. [49] shows a closely related orientation progression; stimulus phase was unavailable for these recordings, so this panel is coloured by orientation alone. Black circles mark projection overlaps within each embedding.

We then applied the same visualisation to ask whether biological salt-and-pepper cortex produced a related population signature. A selected high-population-activity subset of mouse V1 responses from the publicly available Stringer dataset formed another smooth, folded surface with a clear progression of stimulus orientation that closely resembled the model response embeddings (Fig 8d) [49]. The qualitative correspondence across these projections suggests an orderly population-level signature that remains visible even when the corresponding spatial map does not.

Discussion

We asked whether the large domains and pinwheels associated with cortical self-organisation are necessary products of that process or only one regime of it, and our results support the second possibility. Shrinking the classical GCAL interactions preserved fidelity and representational complexity at a shared functional reference point,

but made settling progressively less robust. Micro-GCAL recovered robust settling while
organising the sheet at a much finer scale, and this organisation survived in population
responses even when local positional disorder masked its spatial pattern.

The central distinction is that micro-GCAL is not simply macro-GCAL compressed
into a smaller map. When the models were matched for fidelity, representational
complexity, and settling robustness, the spacing between spatial domains, Λ , was about
3.6-fold smaller in micro-GCAL than in macro-GCAL. Because domain area scales with
the square of this spacing, a characteristic micro-domain occupied approximately 7–8%
of the area of a macro-domain, making it over an order of magnitude smaller. Selective
cross-domain excitation therefore shifted self-organisation onto a distinct spatial scaling
relation. The resulting maps developed sharply tuned receptive fields, broad orientation
coverage, smooth retinotopy, structured like-to-like interactions, and organised
responses to natural images, which show that the self-organising process occurred
despite producing very small domains.

This separation between spatial regimes explains why the same positional disorder
affected the two maps so differently. The displacement chosen to make macro-GCAL
visually comparable with the experimental tree-shrew map moved units by an average of
only two model positions, comparable to a distance of $50\mu\text{m}$ in real cortex. It left the
macro-domain pattern recognisable but spanned a much larger fraction of the distance
between micro-domains, masking their periodic arrangement in cortical coordinates and
producing a squirrel-like salt-and-pepper appearance. Dense tuning measurements still
retained weak structure over short distances, consistent with fine-scale clustering in
mouse V1 [28,30], whereas no clear periodic peak was recovered when a sparse sample
of the learned lateral weights was analysed with the method of Van Hooser et al. [33].
Sparse connectivity is therefore a less sensitive spatial readout of micro-domains than
dense cellular tuning. Tangential dispersion during cortical development provides a
plausible source of this positional mixing [50]. Studies showing related tuning and
preferential synaptic connectivity among clonally related visual cortical neurons further
suggest how fine-scale organisation could develop within restricted ensembles and later
become difficult to read from their positions on the cortical surface [51–53].

In the cortex of species with smooth macro-domain maps, domain size scales
modestly but significantly with cortical size [25,54]. If we assume this to be true for

species with hypothetical micro-domains too, the best place to find them is the cortex of larger rodents, where positional disorder might not have erased all detectable signatures. Two existing experiments in gray squirrel and agouti have attempted to find smooth maps, but were tuned for substantially larger periods. Van Hooser et al. [47] examined gray-squirrel V1 with measurements and analyses aimed at the macroscopic periods found in species with clear orientation maps. Tree-shrew V1 has an orientation-map period of approximately $600\text{ }\mu\text{m}$ [24]. Applying the 3.6-fold linear scale difference measured here predicts a squirrel micro-domain period of about $600/3.6 \approx 170\text{ }\mu\text{m}$, equivalent to approximately 6 cycles/mm. The gray-squirrel study therefore provided a strong test for a tree-shrew-like macro-domain map but did not target the finer period predicted here. Likewise, in agouti, Ferreiro et al. [31] combined optical imaging and electrophysiology but reported that their sampling could not exclude organisation below approximately $250\text{ }\mu\text{m}$, which includes the predicted micro-domain range. Dense tuning maps spanning several candidate $170\text{ }\mu\text{m}$ periods and analysed at the cellular level would provide a more direct test of this prediction.

The population-response analysis provides a second route to the organisation hidden by displacement. Full-phase gratings form a Klein bottle through their joint angle-phase identification, and the stimulus, macro-GCAL, and micro-GCAL UMAPs all formed similarly smooth surfaces with an orderly progression of grating angle. The model response embeddings were therefore compatible with Klein-bottle structure. The mouse recordings supplied orientation but not phase, and their embedding preserved the corresponding orientation progression in biological salt-and-pepper cortex. Recordings that sample angle and phase together in large populations would allow a formal topological comparison across macro-domain and salt-and-pepper species.

The existence of two regimes with comparable coding performance and robustness raises the question of when evolution may favour one over the other. The broader premise that spatial map patterns should be explained through their functional and wiring consequences rather than treated as ends in themselves follows Wilson and Bednar [55] (see also [56]). Mendini and Wilson recently developed this into a comparative account based on correlation-pool size, defined as the number of neurons with which a given neuron is significantly correlated [54]. When this pool is small, relatively uniform connectivity can link correlated neurons at modest cost. As the pool

grows, maintaining those links directly over long distances becomes increasingly
expensive. Grouping correlated neurons into larger local domains then creates pooling
units that cost more to wire internally but allow long-range coupling to become sparser.
Macro-domain maps emerge when this saving in long-range wiring exceeds the
additional local cost, making visible spatial patterning a consequence of the more
economical wiring regime rather than a direct target of selection.

Because macro-GCAL and micro-GCAL produce spatially different maps that are
matched on fidelity, complexity, and robustness, their different distributions of lateral
coupling provide a possible basis for testing this hypothesis. In the account proposed by
Mendini and Wilson, these organisations originate in layer 4 and are inherited by layer
2/3, where horizontal connections span much longer distances. A layer-2/3 network
inheriting micro-domains would require fewer connections to maintain each small local
domain, but might need denser long-range coupling because each domain pools
responses from fewer neurons. A network inheriting macro-domains would require more
connections to maintain its larger local domains, but the larger potential pool size may
allow long-range connectivity to be sparser at similar wiring cost. Adding a distinct
layer 2/3 to both models and comparing the total wiring required to maintain
functional equivalence is a logical next step.

Salt-and-pepper patterns in the visual cortices of rodents and other species are often
considered as contradictory to the universality of self-organisation as a theory of cortical
development. Our results support a different interpretation, in which they may be just
the surface-level appearance of a similar self-organisation, consisting of micro-domains
that are otherwise highly structured by selective inter-neuronal coupling. Despite the
apparent disorder, that process may similarly construct well-tuned receptive fields,
robust stimulus-evoked intra-cortical dynamics, and orderly population responses, and
be favoured by natural selection under certain constraints.

Materials and methods

Model overview and geometry

We modelled a retino-geniculo-cortical pathway with a recurrent V1 sheet, following the GCAL model [4]; and the wider LISSOM family of models [57]. Retinal preprocessing produced an ON-centre channel that passed through local contrast gain control. Each V1 unit sampled a local LGN patch and interacted with nearby V1 units through recurrent lateral fields.

Let $n_{\text{lgn}} = 24$ denote the side length of the square afferent sheet and $N_{\text{lgn}} = n_{\text{lgn}}^2$ its number of units. In the fidelity-complexity analysis, V1 side lengths were $n_{\text{ctx}} \in \{24, 48, 72\}$, giving $N_{\text{ctx}} = n_{\text{ctx}}^2$, V1/LGN side-length ratios of 1, 2, and 3, and expansion factors $\Delta = N_{\text{ctx}}/N_{\text{lgn}} \in \{1, 4, 9\}$. In the matched robustness sweeps, cortical side length varied with interaction scale from approximately 10 to 75 units. We defined the continuous short-range excitatory neighbourhood size as

$$N_{\text{lat}} = \pi r_{\text{sre}}^2, \quad (1)$$

and compared conditions using $R = N_{\text{ctx}}/N_{\text{lat}}$. Circular interaction masks were linearly tapered across their outer lattice spacing, and lateral interactions used zero-padded boundaries. Three independently initialised and trained networks were used for every condition in the replicated sweeps. Unless stated otherwise, error bars show ± 2 standard deviations across these three runs and describe run-to-run variation.

Stimulus ensemble and preprocessing

Natural-image stimuli were drawn from ImageNet [58] and converted to greyscale. Preprocessing was applied once to each source image. A difference-of-Gaussians response was calculated as

$$r = G_{1.2} * I - G_{4.8} * I, \quad (2)$$

where G_{σ} is a Gaussian kernel with standard deviation σ , I is image intensity, and $*$ denotes convolution. The rectified response $x = [r]_+$ was transformed by local contrast

gain control,

$$a_{\text{lgn}} = \tanh\left(0.3 \frac{x}{0.01 + G_4 * x}\right). \quad (3)$$

Each presentation used a randomly located crop drawn wholly from valid image pixels. A rotation was sampled uniformly between -180° and 180° and applied with bilinear interpolation to a source patch large enough that the final 24×24 crop did not intersect rotation padding. For an image too small to support a sampled angle, the angle was resampled. Presentations whose mean ON activity did not exceed 0.15 were discarded. Networks and their reconstruction decoders were trained for two passes through the indexed image set. A fixed, separately seeded set of 10,000 accepted transformations was then generated for post-training evaluation. These test presentations did not update cortical or decoder weights; their source images could also have contributed different training transformations.

Cortical dynamics and settling

In response to each stimulus presentation, cortical activity was updated for $T = 30$ recurrent steps. Each V1 unit i received an afferent drive and one drive from each lateral field, through weighted connections (W);

$$h_{i,\text{aff}} = \sum_j W_{ij,\text{aff}} a_{j,\text{lgn}}, \quad h_{i,*}(t) = \sum_j W_{ij,*} a_{j,\text{ctx}}(t), \quad * = \text{sre, lri, cde}. \quad (4)$$

SRE denotes short-range excitation, LRI longer-range inhibition, and CDE the additional cross-domain excitation present only in micro-GCAL. Afferent weights connected each V1 unit to a local LGN patch, while lateral weights were restricted by the radii described below.

The lateral drive was

$$h_{i,\text{lat}}(t) = \kappa_{\text{sre}} h_{i,\text{sre}}(t) + \kappa_{\text{cde}} h_{i,\text{cde}}(t) - h_{i,\text{lri}}(t). \quad (5)$$

We used $(\kappa_{\text{sre}}, \kappa_{\text{cde}}) = (1, 0)$ in macro-GCAL and $(0.25, 0.7)$ in micro-GCAL. The CDE-absent comparison used the scaled macro-GCAL control defined below: its SRE and LRI radii matched those of the corresponding micro-GCAL map, its sheet size preserved the macro-GCAL reference ratio, and its recurrent excitation was confined to

SRE. The same afferent and lateral gain targets were used in both conditions. Separate gains controlled the afferent and lateral contributions,

$$u_i(t) = \gamma_{\text{aff}} (h_{i,\text{aff}} - b_{\text{aff}}) + \gamma_{\text{lat}} h_{i,\text{lat}}(t),$$

where $b_{\text{aff}} = 0.3$ was a fixed afferent baseline applied to make the homeostatic thresholds converge faster. Activity was updated directly as

$$a_{i,\text{ctx}}(t+1) = \tanh[\max(0, u_i(t) + \eta_i(t) - \theta_i)]. \quad (6)$$

Here θ_i was a unit-specific adaptive threshold and $\eta_i(t)$ was zero except during the robustness analysis.

Gain calibration

Separate gains controlled the afferent and lateral contributions. Let

$$d_{i,\text{aff}} = \gamma_{\text{aff}} (h_{i,\text{aff}} - b_{\text{aff}}), \quad d_{i,\text{lat}} = \gamma_{\text{lat}} h_{i,\text{lat}}(T), \quad (7)$$

where $b_{\text{aff}} = 0.3$. After each settled response, unit-specific running estimates of the positive contribution from each branch $x \in \{\text{aff}, \text{lat}\}$ were updated as

$$m_{i,x} \leftarrow (1 - g_i)m_{i,x} + g_i[d_{i,x}]_+, \quad g_i = 10\alpha_\gamma a_{i,\text{ctx}}(T). \quad (8)$$

Writing $c_x = N_{\text{ctx}}^{-1} \sum_i m_{i,x}$, the corresponding gain was updated according to

$$\gamma_x \leftarrow \gamma_x - \alpha_\gamma \frac{c_x - \mu_x}{\mu_x} \quad (9)$$

Both architectures used $\mu_{\text{aff}} = \mu_{\text{lat}} = 0.13$. This kept the target balance between total afferent and lateral drive fixed while allowing their spatial organisation to differ between architectures.

Homeostatic threshold

427

For each V1 unit, a running mean of settled activity was updated as

428

$$\bar{a}_i \leftarrow (1 - \alpha_\theta)\bar{a}_i + \alpha_\theta a_{i,\text{ctx}}(T). \quad (10)$$

The unit-specific response threshold was then adjusted towards the target mean activity

429

$\mu_\theta = 0.05$,

430

$$\theta_i \leftarrow \theta_i + \alpha_\theta \frac{\bar{a}_i - \mu_\theta}{\mu_\theta} \quad (11)$$

The gain and threshold adaptation rates were equal, $\alpha_\gamma = \alpha_\theta$, at each training presentation.

431

432

Plasticity

433

Afferent weights, longer-range inhibitory weights, and micro-GCAL CDE weights were learned through Hebbian plasticity,

434

435

$$\Delta W_{ij,*} = \alpha_H y_i x_j, \quad (12)$$

where x_j and y_i were the pre- and postsynaptic settled activities. SRE was fixed rather than learned. After every update, incoming weights within each permitted connection field were rectified and divided by their spatial mean. The learning rate was exponentially decreased by a factor of 10 throughout learning.

436

437

438

439

Lateral interaction geometry and parametrisation

440

We used the SRE radius, r_{sre} , as the reference spatial scale. In both architectures, the longer-range inhibitory radius was

441

442

$$r_{\text{lri}} = 3r_{\text{sre}}. \quad (13)$$

Macro-GCAL contained SRE and LRI only. Micro-GCAL additionally contained cross-domain excitation (CDE) with radius

443

444

$$r_{\text{cde}} = 4.5r_{\text{sre}}. \quad (14)$$

These relative radii and the recurrent coefficients were held fixed when the overall interaction scale was varied.

The fidelity–complexity sweep used 12 SRE scales spanning $r_{\text{sre}} \approx 0.67$ –5 lattice spacings. The fixed 60×60 maps used $r_{\text{sre}} = 4$ for macro-GCAL and $r_{\text{sre}} \approx 1.48$ for micro-GCAL. The CDE-absent comparison was a 22×22 macro-GCAL map with the same SRE and LRI radii as the 60×60 micro-GCAL map. Positional-permutation and connectivity analyses used 90×90 maps with $r_{\text{sre}} = 6$ for macro-GCAL and $r_{\text{sre}} \approx 2.22$ for micro-GCAL.

Reconstruction fidelity and representational complexity

To quantify how much afferent structure was recoverable from settled V1 activity, we trained a three-layer multilayer perceptron to reconstruct the 24×24 LGN input from the cortical response. The decoder was trained with mean-squared error alongside each map but did not feed back into cortical development.

After development, cortical and decoder weights were frozen. Reconstruction and dimensionality were evaluated using the same fixed set of $K = 10,000$ post-training presentations described above. Reconstruction accuracy was the mean cosine similarity

$$A = \frac{1}{K} \sum_{k=1}^K \text{cosim}(\mathbf{a}_{\text{lgn}}^{(k)}, \phi(\mathbf{a}_{\text{ctx}}^{(k)})), \quad (15)$$

where ϕ denotes the trained decoder.

The reconstruction baseline was calculated from three null decoders that received the same constant input for every stimulus. They were trained and evaluated on the same presentation sets as the model decoders. Their mean cosine similarity gave $A_{\text{null}} \approx 0.63$. Fidelity was

$$F = \frac{A - A_{\text{null}}}{1 - A_{\text{null}}}. \quad (16)$$

Complexity was measured by PCA. We defined D_{ctx} and D_{lgn} as the numbers of components required to explain 95% of the variance in settled V1 responses and LGN ON responses, respectively. Under valid random cropping, the shared test set gave

$D_{\text{Ign}} = 87$, and relative representational complexity was

$$C = \frac{D_{\text{ctx}}}{D_{\text{Ign}}} = \frac{D_{\text{ctx}}}{87}. \quad (17)$$

Values of $C < 1$ indicate that recurrent cortical settling reduced dimensionality relative to the afferent representation. Micro-GCAL was matched to the fidelity and complexity of macro-GCAL at C_{curv} .

Reference-point fitting

To characterise the macro-GCAL fidelity–complexity trade-off (Fig 2), we varied N_{lat} from approximately 1 to 80 at V1/LGN side-length ratios of 1, 2, and 3. At every parameter point, fidelity and complexity were averaged across three independently trained networks.

For each sheet size, fidelity and complexity were fitted separately as functions of N_{lat} using a shifted-power function,

$$y(N_{\text{lat}}) = A \left(1 + \frac{N_{\text{lat}} - N_{\text{min}}}{\tau} \right)^{-p}, \quad (18)$$

where N_{min} was the smallest tested neighbourhood. The second simulated point is shown but was not used for fitting the fidelity and complexity functions, as it deviates from the general trend significantly, likely due to discretisation effects at small interaction sizes.

Neighbourhood size was then eliminated analytically between the fitted fidelity and complexity functions for each sheet size, and the three resulting curves were averaged pointwise. No additional function was fitted directly to the fidelity–complexity observations.

The reference point was defined from this mean analytical curve. Fidelity was divided by its fitted plateau to give q , and complexity was expressed as $u = \log(C/C_{50})$, where C_{50} is the complexity at half of the fitted plateau. Signed curvature was

$$\kappa(u) = \frac{q''(u)}{[1 + q'(u)^2]^{3/2}}, \quad (19)$$

and C_{curv} was the point of strongest negative curvature, corresponding to the bend towards saturation.

Mapping C_{curv} through the sheet-size-specific fits gave $R_{\text{macro}}^* = N_{\text{ctx}}/N_{\text{lat}} \approx 72$. For micro-GCAL, the complete SRE–LRI–CDE motif was scaled with its relative radii and recurrent coefficients fixed, and the scale matching macro-GCAL fidelity and complexity at C_{curv} gave $R_{\text{micro}}^* \approx 523$.

Noise injection, stability and robustness metrics

Robustness was measured by injecting additive temporally correlated Gaussian noise into the cortical drive during settling. For each unit,

$$n_i(t) = \beta n_i(t-1) + (1-\beta)\xi_i(t), \quad \eta_i(t) = \sigma_{\text{noise}} n_i(t), \quad (20)$$

where $\xi_i(t)$ was independent Gaussian noise, $\beta = 0.8$, and 15 values of σ_{noise} were sampled uniformly from 0 to 0.1. No learning occurred during these measurements.

The replicated scale sweeps reused the same 10,000 test presentations employed for fidelity and complexity at every noise intensity. For a fixed stimulus, stability was

$$\text{stability} = \text{cosim}(\mathbf{a}_{\text{ctx}}, \tilde{\mathbf{a}}_{\text{ctx}}), \quad (21)$$

where $\mathbf{a}_{\text{ctx}}$ and $\tilde{\mathbf{a}}_{\text{ctx}}$ are the clean and noisy settled states. Similarity was averaged across presentations. Robustness was the noise intensity at which mean stability fell to 0.95, interpolated between adjacent tested intensities.

The fixed-geometry CDE comparison used 1,000 presentations and the same stability definition.

Orientation selectivity, map estimation, and domain spacing

Orientation preference was estimated from settled responses to sinusoidal gratings. We presented 18 orientations uniformly spanning $[0, \pi)$, each at eight phases uniformly spanning $[0, 2\pi)$, with spatial frequency 0.05 cycles per pixel. Responses were averaged across phase. For phase-averaged response r_k at orientation ϑ_k , orientation selectivity

was the axial vector strength

514

$$\text{OSI} = \frac{|\sum_k r_k e^{2i\vartheta_k}|}{\sum_k r_k}, \quad (22)$$

and preferred orientation was half the argument of the same complex sum. No learning occurred during grating presentation.

515

516

Domain spacing Λ was estimated from the resulting orientation-preference map $\psi(x)$. Orientations were represented by the axial field

517

518

$$z(x) = e^{2i\psi(x)}. \quad (23)$$

The spatial mean of $\text{Re } z(x)$ was subtracted, its two-dimensional Fourier-amplitude spectrum was calculated, and the dominant non-zero radial frequency f^* was identified from the ring profile. Domain spacing was $\Lambda = 1/f^*$.

519

520

521

Post-learning positional permutation

522

Positional-displacement analyses used the 90×90 macro- and micro-GCAL maps described above. After learning, units were swapped with partners sampled from a two-dimensional Gaussian centred on their lattice positions. The same displacement was applied to both architectures. This changed spatial organisation while preserving each unit's response properties and learned connections.

523

524

525

526

527

We tested Gaussian standard deviations $\sigma \in [0, 2]$ lattice units. Spatial tuning analyses used the central 45×45 region. The illustrative condition, $\sigma = 1.5$, produced a mean displacement of approximately two lattice positions, or $50 \mu\text{m}$ under the 45-units/mm calibration.

528

529

530

531

Tuning-based periodicity measures

532

To quantify spatial periodicity in tuning, we computed two distance-dependent summaries from the axial orientation field. We radially averaged the normalised two-dimensional autocorrelation of the mean-centred field $z(x) = e^{i2\psi(x)}$ and the absolute circular difference in preferred orientation across unit pairs. These measures were used in Fig 7c to track how periodic signatures changed with displacement.

533

534

535

536

537

Connectivity-field analysis

Connectivity analysis used learned Hebbian fields from each 90×90 map. Connections exceeding 1.5x the global mean weight were eligible for sampling, from which we retained up to 25% of the full field to form a binary sparse field. This post-development sampling did not alter model dynamics or learning.

Sparse fields were spatially windowed, aligned to their principal axis, and Fourier transformed. Their normalised central Fourier-amplitude profiles were averaged across target units. Ideal and displaced fields were processed identically.

Orientation-phase response analysis

We defined a sinusoidal grating at wave-vector angle α and spatial phase ϕ by

$$g_{\alpha,\phi}(x,y) = \cos(2\pi\nu[x \cos \alpha + y \sin \alpha] + \phi), \quad (24)$$

at fixed spatial frequency ν and contrast. These images satisfy

$$g_{\alpha+\pi,\phi} = g_{\alpha,-\phi}, \quad (25)$$

which gives the space of distinct full-phase gratings its Klein-bottle identification [48]. The stimulus generator sampled both parameters over $[0, 2\pi)$.

We generated 3,831 gratings with random angles and phases and collected the settled responses of the 60×60 macro- and micro-GCAL maps. Stimulus and response matrices were reduced separately to 50 principal components and embedded independently in three dimensions with UMAP ($n_{\text{neighbors}} = 15$, $\text{min_dist} = 0.5$; [59]).

For the empirical comparison, we used recording 1 (TX36; file `static_sin_rand_TX36_2019_10_21_1.npy`) from the oriented-stimulus calcium-imaging data from mouse V1 [49]. Each neuron's activity was z-scored across trials. For trial selection only, each neuron's mean response within each of 24 orientation bins was subtracted, and the resulting residual activity was averaged across all 18,802 recorded neurons for each trial. Orientation bins were retained only when they contained at least 15 trials. Trials at or above the median population-average residual across valid trials were designated the high-population-activity subset. All 3,831 trials were valid, yielding

1,916 selected trials. For the population-response embedding, we used the original
z-scored responses before orientation-bin mean subtraction. Responses from a random
sample of 6,000 neurons were reduced to 50 principal components before UMAP, so the
projection shown in Fig 8d contained 1,916 observations. Trial phase was unavailable
for these recordings.

Artificial intelligence tools

AI tools (ChatGPT Pro) were used to generate the animal icons included in Figs 1, 4,
and 7. The generated images were reviewed and edited by the authors and were used
solely as illustrative elements.

Ethics statement

No new animal experiments were performed in this study. The empirical analyses used
only previously published, publicly available mouse primary visual cortex recordings.
The animal procedures and ethical approvals associated with the original collection of
these data are described in the original publication. The present study comprised
secondary computational analysis of existing data and involved no new contact with
animals or animal tissue; therefore, no additional animal ethics approval was sought for
this work.

Data availability

All author-generated code, model configuration files, random seeds, and numerical
source data underlying the figures are available at
<https://github.com/nicolamendini/microdomains>.

The publicly available mouse primary visual cortex recordings analysed in this study
are available from Figshare (doi:10.25378/janelia.8279387.v3). Natural-image stimuli
were obtained from ImageNet [58] under its standard conditions of access. The
identifiers and random seeds required to reproduce the sampled image patches are
provided with the analysis code.

References

1. von der Malsburg C. Self-organization of orientation sensitive cells in the striate cortex. *Kybernetik*. 1973 Dec;14(2):85–100.
2. Sirosh J, Miikkulainen R. Cooperative self-organization of afferent and lateral connections in cortical maps. *Biological Cybernetics*. 1994 May;71(1):65–78. Available from: <https://doi.org/10.1007/BF00198912>.
3. Miikkulainen R, Bednar JA, Choe Y, Sirosh J. *Computational Maps in the Visual Cortex*. Springer. 2005;.
4. Stevens JLR, Law JS, Antolik J, Bednar JA. Mechanisms for Stable, Robust, and Adaptive Development of Orientation Maps in the Primary Visual Cortex. *The Journal of Neuroscience*. 2013 Oct;33(40):15747–15766.
5. Kaschube M. Neural maps versus salt-and-pepper organization in visual cortex. *Current Opinion in Neurobiology*. 2014 Feb;24:95–102.
6. Rockland KS. A reticular pattern of intrinsic connections in primate area V2 (area 18). *The Journal of Comparative Neurology*. 1985 May;235(4):467–478.
7. Levitt JB, Yoshioka T, Lund JS. Intrinsic cortical connections in macaque visual area V2: evidence for interaction between different functional streams. *The Journal of Comparative Neurology*. 1994 Apr;342(4):551–570.
8. Gilbert CD, Wiesel TN. Columnar specificity of intrinsic horizontal and corticocortical connections in cat visual cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*. 1989 Jul;9(7):2432–2442.
9. Malach R, Amir Y, Harel M, Grinvald A. Relationship between intrinsic connections and functional architecture revealed by optical imaging and in vivo targeted biocytin injections in primate striate cortex. *Proceedings of the National Academy of Sciences*. 1993 Nov;90(22):10469–10473.
10. Yoshioka T, Blasdel GG, Levitt JB, Lund JS. Relation between Patterns of Intrinsic Lateral Connectivity, Ocular Dominance, and Cytochrome

- Oxidase-Reactive Regions in Macaque Monkey Striate Cortex. *Cerebral Cortex*. 1996 Mar;6(2):297–310.
11. Bosking WH, Zhang Y, Schofield B, Fitzpatrick D. Orientation Selectivity and the Arrangement of Horizontal Connections in Tree Shrew Striate Cortex. *Journal of Neuroscience*. 1997 Mar;17(6):2112–2127.
 12. Rumberger A, Tyler CJ, Lund JS. Intra- and inter-areal connections between the primary visual cortex V1 and the area immediately surrounding V1 in the rat. *Neuroscience*. 2001 Jan;102(1):35–52.
 13. Sincich LC, Blasdel GG. Oriented Axon Projections in Primary Visual Cortex of the Monkey. *The Journal of Neuroscience*. 2001 Jun;21(12):4416–4426.
 14. Stettler DD, Das A, Bennett J, Gilbert CD. Lateral Connectivity and Contextual Interactions in Macaque Primary Visual Cortex. *Neuron*. 2002 Nov;36(4):739–750.
 15. Koulakov AA, Chklovskii DB. Orientation Preference Patterns in Mammalian Visual Cortex: A Wire Length Minimization Approach. *Neuron*. 2001 Feb;29(2):519–527.
 16. Chklovskii DB, Koulakov AA. Maps in the brain: what can we learn from them? *Annual Review of Neuroscience*. 2004;27:369–392.
 17. Hubel DH, Wiesel TN. Sequence regularity and geometry of orientation columns in the monkey striate cortex. *The Journal of Comparative Neurology*. 1974 Dec;158(3):267–293.
 18. Bonhoeffer T, Grinvald A. Iso-orientation domains in cat visual cortex are arranged in pinwheel-like patterns. *Nature*. 1991 Oct;353(6343):429–431.
 19. Swindale N, Matsubara J, Cynader M. Surface organization of orientation and direction selectivity in cat area 18. *The Journal of Neuroscience*. 1987 May;7(5):1414–1427.
 20. Issa NP, Trepel C, Stryker MP. Spatial Frequency Maps in Cat Visual Cortex. *Journal of Neuroscience*. 2000 Nov;20(22):8504–8514.

21. Blasdel GG. Orientation selectivity, preference, and continuity in monkey striate cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*. 1992 Aug;12(8):3139–3161.
22. Obermayer K, Blasdel GG. Geometry of orientation and ocular dominance columns in monkey striate cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*. 1993 Oct;13(10):4114–4129.
23. Wolf F. Symmetry, Multistability, and Long-Range Interactions in Brain Development. *Physical Review Letters*. 2005 Nov;95(20):208701.
24. Kaschube M, Schnabel M, Löwel S, Coppola DM, White LE, Wolf F. Universality in the evolution of orientation columns in the visual cortex. *Science*. 2010 Nov;330(6007):1113–1116.
25. Ho CLA, Zimmermann R, Flórez Weidinger JD, Prsa M, Schottdorf M, Merlin S, et al. Orientation Preference Maps in *Microcebus murinus* Reveal Size-Invariant Design Principles in Primate Visual Cortex. *Current Biology*. 2021 Feb;31(4):733–741.e7.
26. Swindale NV. Coverage and the design of striate cortex. *Biological Cybernetics*. 1991 Oct;65(6):415–424.
27. Ohki K, Chung S, Ch'ng YH, Kara P, Reid RC. Functional imaging with cellular resolution reveals precise micro-architecture in visual cortex. *Nature*. 2005 Feb;433(7026):597–603.
28. Bonin V, Histed MH, Yurgenson S, Reid RC. Local Diversity and Fine-Scale Organization of Receptive Fields in Mouse Visual Cortex. *Journal of Neuroscience*. 2011 Dec;31(50):18506–18521.
29. Han S, Yang W, Yuste R. Two-Color Volumetric Imaging of Neuronal Activity of Cortical Columns. *Cell reports*. 2019 May;27(7):2229–2240.e4.
30. Ringach DL, Mineault PJ, Tring E, Olivas ND, Garcia-Junco-Clemente P, Trachtenberg JT. Spatial clustering of tuning in mouse primary visual cortex. *Nature Communications*. 2016 Aug;7(1):12270. Available from: <https://www.nature.com/articles/ncomms12270>.

31. Ferreiro DN, Conde-Ocazonez SA, Patriota JHN, Souza LC, Oliveira MF, Wolf F, et al. Spatial clustering of orientation preference in primary visual cortex of the large rodent agouti. *iScience*. 2021 Jan;24(1).
32. Yu Y Peijia Yang, Gozel O, Oldenburg I, Dipoppa M, Rossi LF, Miller K, et al. Circuit-Based Understanding of Fine Spatial Scale Clustering of Orientation Tuning in Mouse Visual Cortex. *bioRxiv*. 2025 Feb;.
33. Van Hooser SD, Heimel JA, Chung S, Nelson SB. Lack of Patchy Horizontal Connectivity in Primary Visual Cortex of a Mammal without Orientation Maps. *Journal of Neuroscience*. 2006 Jul;26(29):7680–7692.
34. Burkhalter A, Charles V. Organization of local axon collaterals of efferent projection neurons in rat visual cortex. *The Journal of Comparative Neurology*. 1990 Dec;302(4):920–934.
35. Soodak RE. The retinal ganglion cell mosaic defines orientation columns in striate cortex. *Proceedings of the National Academy of Sciences*. 1987 Jun;84(11):3936–3940. Available from:
<https://www.pnas.org/doi/abs/10.1073/pnas.84.11.3936>.
36. Soodak RE. Two-dimensional modeling of visual receptive fields using Gaussian subunits. *Proceedings of the National Academy of Sciences*. 1986 Dec;83(23):9259–9263. Available from:
<https://www.pnas.org/doi/abs/10.1073/pnas.83.23.9259>.
37. Paik SB. Developmental models of functional maps in cortex. *Biomedical Engineering Letters*. 2013 Dec;3(4):223–225. Available from:
<https://doi.org/10.1007/s13534-013-0115-x>.
38. Paik SB, Ringach DL. Link between orientation and retinotopic maps in primary visual cortex. *Proceedings of the National Academy of Sciences*. 2012 May;109(18):7091–7096. Available from:
<https://www.pnas.org/doi/full/10.1073/pnas.1118926109>.

39. Paik SB, Ringach DL. Retinal origin of orientation maps in visual cortex. *Nature Neuroscience*. 2011 Jul;14(7):919–925. Available from:
<https://www.nature.com/articles/nn.2824>.
40. Schotttdorf M, Keil W, Coppola D, White LE, Wolf F. Random Wiring, Ganglion Cell Mosaics, and the Functional Architecture of the Visual Cortex. *PLoS computational biology*. 2015 Nov;11(11):e1004602.
41. Weigand M, Sartori F, Cuntz H. Universal transition from unstructured to structured neural maps. *Proceedings of the National Academy of Sciences*. 2017 May;114(20):E4057–E4064. Available from:
<https://www.pnas.org/doi/10.1073/pnas.1616163114>.
42. Jang J, Song M, Paik SB. Retino-Cortical Mapping Ratio Predicts Columnar and Salt-and-Pepper Organization in Mammalian Visual Cortex. *Cell Reports*. 2020 Mar;30(10):3270–3279.e3.
43. Corey J, Scholl B. Cortical Selectivity through Random Connectivity. *Journal of Neuroscience*. 2012 Jul;32(30):10103–10104. Available from:
<https://www.jneurosci.org/content/32/30/10103>.
44. Barlow HB. Possible Principles Underlying the Transformations of Sensory Messages. In: Rosenblith WA, editor. *Sensory Communication*. The MIT Press; 2012. p. 0.
45. Olshausen BA, Field DJ. Sparse coding of sensory inputs. *Current Opinion in Neurobiology*. 2004 Aug;14(4):481–487.
46. Sachdeva V, Mora T, Walczak AM, Palmer SE. Optimal prediction with resource constraints using the information bottleneck. *PLoS Computational Biology*. 2021 Mar;17(3):e1008743. Available from:
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7971903/>.
47. Hooser SDV, Heimel JAF, Chung S, Nelson SB, Toth LJ. Orientation Selectivity without Orientation Maps in Visual Cortex of a Highly Visual Mammal. *Journal of Neuroscience*. 2005 Jan;25(1):19–28.

48. Swindale NV. Visual cortex: Looking into a Klein bottle. *Current Biology*. 1996 Jul;6(7):776–779. Available from: <https://www.sciencedirect.com/science/article/pii/S0960982202005924>.
49. Stringer C, Michaelos M, Tsyboulski D, Lindo SE, Pachitariu M. High-precision coding in visual cortex. *Cell*. 2021 Apr;184(10):2767–2778.e15.
50. Rakic P. Radial versus tangential migration of neuronal clones in the developing cerebral cortex. *Proceedings of the National Academy of Sciences*. 1995 Dec;92(25):11323–11327. Available from: <https://www.pnas.org/doi/10.1073/pnas.92.25.11323>.
51. Ohtsuki G, Nishiyama M, Yoshida T, Murakami T, Histed M, Lois C, et al. Similarity of Visual Selectivity among Clonally Related Neurons in Visual Cortex. *Neuron*. 2012 Jul;75(1):65–72. Available from: <https://www.sciencedirect.com/science/article/pii/S0896627312005119>.
52. Li Y, Lu H, Cheng Pl, Ge S, Xu H, Shi SH, et al. Clonally Related Visual Cortical Neurons Show Similar Stimulus Feature Selectivity. *Nature*. 2012 May;486(7401):118–121. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3375857/>.
53. Yu YC, Bultje RS, Wang X, Shi SH. Specific synapses develop preferentially among sister excitatory neurons in the neocortex. *Nature*. 2009 Mar;458(7237):501–504. Available from: <https://www.nature.com/articles/nature07722>.
54. Mendini N, Wilson SP. 2.09 - What do species differences in cortical map organization reveal about the evolution of cortical circuitry? In: Kaas JH, Herculano-Houzel S, editors. *Evolution of Nervous Systems (Third Edition)*. third edition ed. Oxford: Academic Press; 2026. p. 213–229. Available from: <https://www.sciencedirect.com/science/article/pii/B9780443273803000506>.
55. Wilson SP, Bednar JA. What, if anything, are topological maps for? *Developmental Neurobiology*. 2015;75(6):667–681.

56. Purves D, Riddle DR, LaMantia AS. Iterated patterns of brain circuitry (or how the cortex gets its spots). *Trends in Neurosciences*. 1992 Oct;15(10):362–368.
57. Miikkulainen R, Bednar JA, Choe Y, Sirosh J. Computations in Visual Maps. In: *Computational Maps in the Visual Cortex*. Springer; 2005. p. 307–324.
58. Deng J, Dong W, Socher R, Li LJ, Li K, Fei-Fei L. ImageNet: A large-scale hierarchical image database. In: *2009 IEEE Conference on Computer Vision and Pattern Recognition*; 2009. p. 248–255. ISSN: 1063-6919. Available from: <https://ieeexplore.ieee.org/document/5206848>.
59. McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction; 2020. Available from: <https://arxiv.org/abs/1802.03426>.